Women in Data Science Worldwide

Beyond seismic design: data-driven modelling of system recovery and resilience

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Context and Motivation

Earthquakes are rare but destructive events that repeatedly threaten our cities and infrastructure. Engineers design buildings to survive them — *how safe is safe enough, and for how long?* Answering this requires three concepts:

Seismic reliability — how safe is a structure. It is the probability that a structure survives repeated earthquakes over its lifetime.

Post-earthquake recovery — what happens *after* damage. A building may survive an earthquake yet remain closed for months. Recovery time matters.

System resilience — the capacity of a structure to absorb disruption, recover swiftly, and maintain acceptable performance over time.

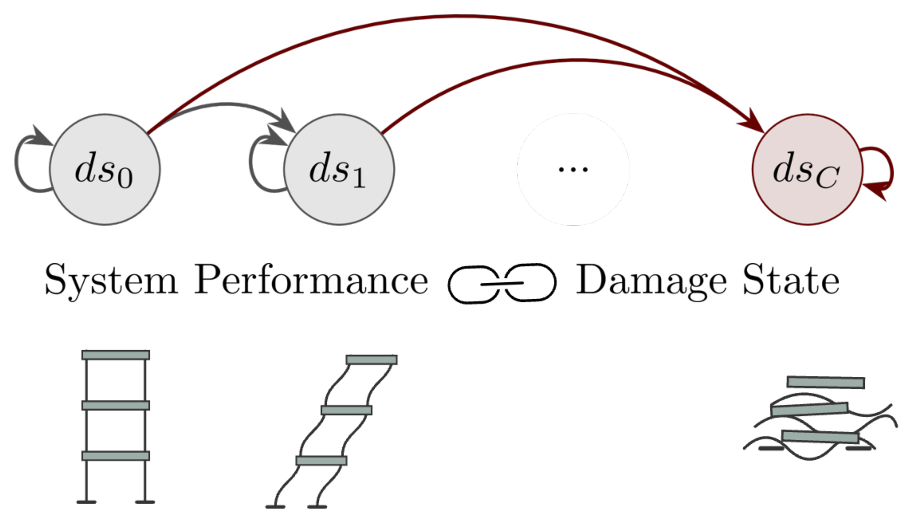
Goal: Can we quantify how recovery shapes structural reliability and resilience over a system's lifetime?

Problem Statement

Buildings experience **sequences** of earthquakes over decades, and between events they are repaired at different recovery times. The key question:

How do repeated damage–recovery cycles shape long-term safety, under uncertainty?

What is the performance (= *damage state, ds*) of the system?



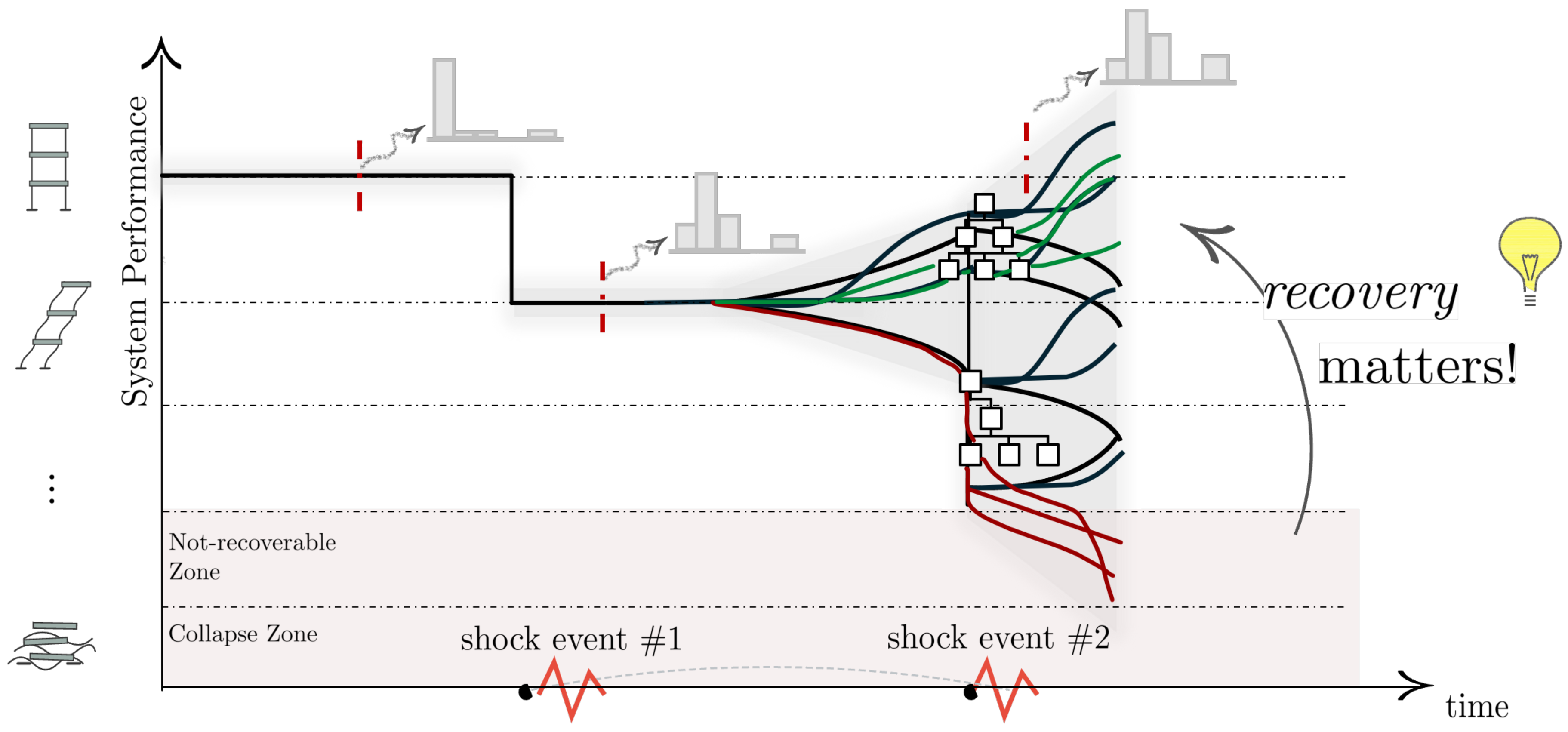
$\pi_i(t) = \mathbb{P}(DS(t) = ds_i)$

probability of occupying each *damage state* at time t

the full distribution evolves with every shock and every recovery

What happens in the aftermath?

Does recovery determine the next failure?

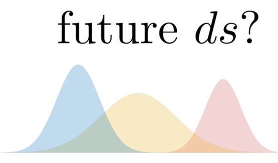


Methodology & Tools

Governing Equation/ Probabilistic Analysis

$$\dot{\boldsymbol{\pi}}(t) = \boldsymbol{\pi}(0) \cdot \boldsymbol{Q}$$

future *ds*? starting *ds* *generator* matrix



Data harvesting and feature extraction

$$\boldsymbol{Q} = \boldsymbol{Q}_{\text{dam}} + \boldsymbol{Q}_{\text{rec}} + \dots$$

seismic damage recovery info

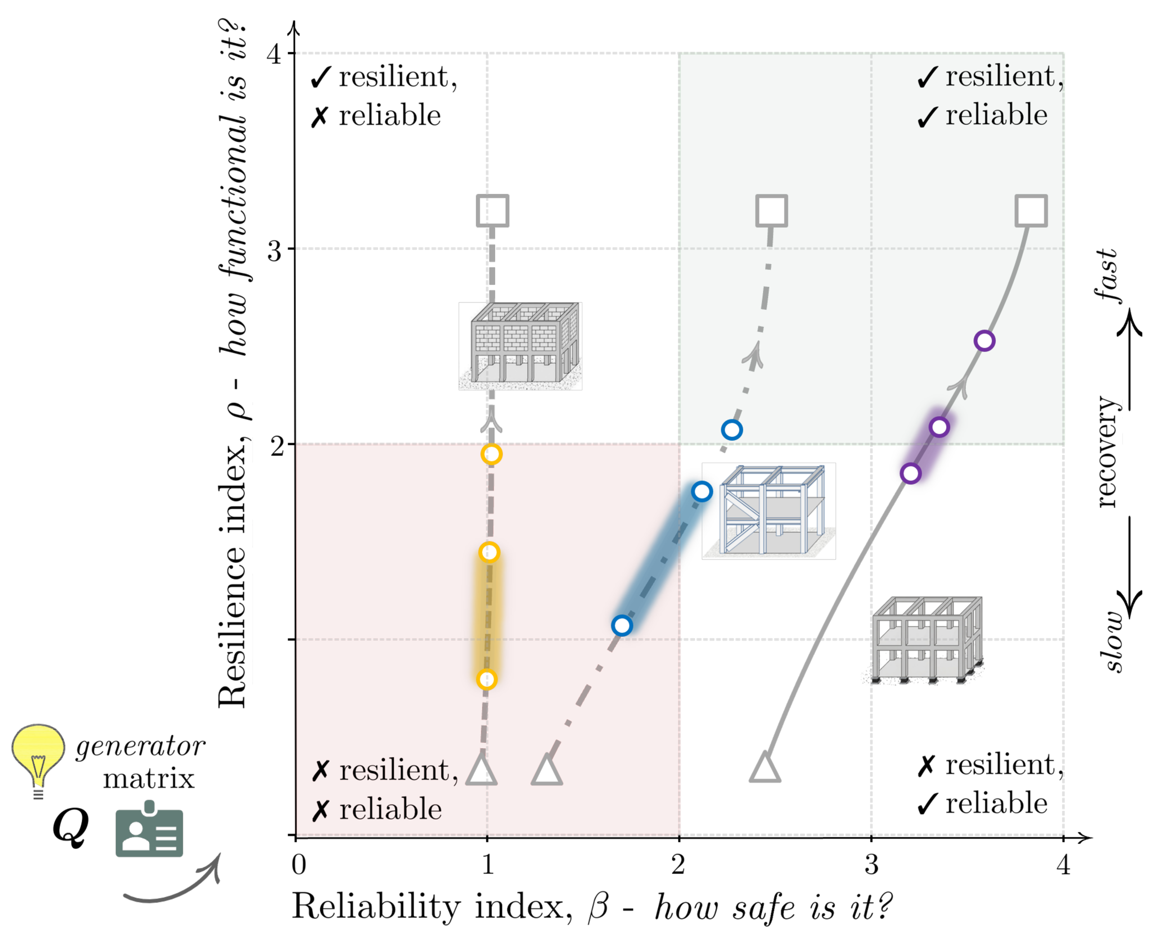
Reliability/Resilience metrics

is it *safe*? is it *functional*?

$\beta_t = -\Phi^{-1} F_{\mathcal{T}}(t, \boldsymbol{Q})$ reliability index

$\rho_t = -\Phi^{-1} \mathcal{R}_0(t, \boldsymbol{Q})$ resilience index

Proof of concept: three systems, three different stories



Recovery time matters—but not always in the same way.

The plot maps three structural systems onto a reliability–resilience plane, where reliability captures long-term safety and resilience reflects how quickly a system returns to full functionality after damage. Each system is evaluated under two recovery scenarios: slow recovery Q_{slow} (▲) and fast recovery Q_{fast} (□).

Faster recovery always improves resilience. Its effect on reliability depends on the system.

For masonry-infilled frames, speeding up repair moves the system almost straight up in the plot: resilience improves, but reliability stays flat.

For braced steel frames, the trajectory begins to tilt. Faster recovery still primarily boosts resilience, but now also nudges reliability — the two metrics start moving together.

For base-isolated buildings, fast recovery pays off on both axes. Already more reliable by design, these systems reach the "resilient and reliable" corner when repair is measured in days rather than months.

The markers on each curve reflect recovery times drawn from engineering codes and post-earthquake reports, grounding each trajectory in realistic expectations for that building class.

Take-Home Message and Future Directions

Beyond seismic damage: *recovery time matters — it shapes not only how fast a system bounces back, but how safe it stays over time.*

Reliability and resilience can — and should — be assessed jointly. The structural typology sets the starting point; recovery dynamics determine how far a system can go.

What's next? Single structures are just the beginning. Real communities are networks — hospitals, bridges, power grids — where local recovery and system-wide resilience are deeply intertwined. Scaling this framework to interconnected infrastructures is the natural next step.

References:

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